

Risk Assessment for Event-Based Surveillance (EBS)

In an Event-Based Surveillance (EBS) system, all events must undergo the process of risk assessment. Risk assessment is the systematic and ongoing process of gathering, evaluating, and documenting information that will form the basis of the actions required to manage and minimise the negative consequences of a serious health event. The process results in assigning a level of risk that an event presents to human, animal, plant, and environment health. Risk assessment should be conducted by health authorities who are responsible for proposing the actions or responses that must be taken to manage and minimise the negative consequences of serious health events.

A risk assessment should be conducted within the first 24 hours of signal verification and should be repeated as new information becomes available until the end of the response to an event. As new information about the situation can arise at any time, the ongoing risk assessment ensures that the appropriate response is triggered, and that it reflects the level of risk the event poses to health. Resources must be set aside to train staff in risk assessment.

Risk assessment should be performed at the lowest administrative level with capacity depending on the magnitude of the event or the capacity of staff across the levels of the sector(s) implementing EBS (e.g., national or intermediate level). The speed with which assessments can be conducted will depend on the relationships the National EBS unit has with local-level health authorities or facilities near the origination of the event. Under the supervision of the national unit, the involvement of local-level health authorities in verification, and where possible a preliminary assessment, will make the system more responsive.

As part of the risk assessment process, you will want to evaluate the level of risk for the following three elements: **hazard**, **exposure**, and **context**.

Hazard Assessment

Hazard assessment is the identification of the characteristics of a health hazard—including possible aetiologies, causes, and/or sources—and the associated adverse health effects. Hazards can include biological, chemical, radiological, and nuclear events.

Helpful information includes:

- Historical clinical outcome information for known events with pathogen/agent lab confirmation
- In all other cases, start with listing possible causes based on:
 1. The initial description of the event;
 2. Available clinical and epidemiological features;
 3. Known burden of diseases in the affected community; and
 4. Type and distribution of existing hazards (e.g., the number and location of chemical plants and the chemicals they use for chemical exposure).

Possible questions to ask

Is this threat unusual or unexpected?

- Is this event new in the country?
- Is this occurring in an unusual or unexpected setting, mode of transmission, and/or population group?
- Are there novel clinical manifestations that result in severe illness, disability, and/or death?

Does the event have possible consequences for human health?

- Is the event associated with high mortality or morbidity?

- E.g., Considering the pathogen (if known), exposed population, and availability of treatment where the event is occurring, will more than 20% of people infected develop severe illness, severe disability, and/or die?
- Does the event have possible consequences for trade or travel?
- Does the event affect livestock/wildlife?
- Are there environmental consequences?

Exposure assessment is the evaluation of the vulnerability of individuals and populations to likely hazards. The key output of the assessment is an estimate of the population that may have been exposed and an estimate of those that may be susceptible.

Helpful information to evaluate exposure includes:

- Historical pathogen/agent information for known events with lab confirmation;
- Mode of transmission/exposure (e.g., direct contact, droplet, sexual, occupational);
- Incubation period (known or suspected);
- Estimation of the potential for transmission (e.g., R_0 basic reproduction number);
- Immune status of the exposed population;
- Disease burden in and distribution of vector or animal hosts/reservoirs for vector-borne and zoonotic diseases;
- Dose (e.g., amount ingested/absorbed/inhaled) and duration of exposure

Possible questions to ask

What is the probability of spread within and beyond the country?

- Does the suspected event have a high potential for spread (e.g., cholera, avian influenza)?
- Is this readily transmitted person-to-person (e.g., airborne)?
- Is exposure geographically widespread or limited to specific sub-groups (e.g., immunodeficient, comorbidities, socially/economically vulnerable, location)?
- Does infection/exposure occur from a low dose of the pathogen or agent?

If an animal event, is there a potential for the pathogen to spread from animals to humans?

- Are treatment or prevention measures available for animals or people?
- Does the event/pathogen cause severe morbidity or mortality in humans?
- Does the agent have the potential for person-to-person transmission?

Context assessment is an evaluation of the environment in which an event is taking place. This may include the physical environment (climate, vegetation, land use, and water systems and sources), the health of the population (nutritional status, disease burden and previous outbreaks), infrastructure (including transportation, clinical, and health systems), and cultural practices and beliefs. Context assessment also considers social, ethical, technical, scientific, economic, environmental, and political factors that can affect the potential severity of the event.

Helpful information includes any social, ethical, technical, scientific, economic, environmental, or political context that could impact the event. For example:

- The physical environment such as climate, vegetation, land use (e.g., farming, industry) and water systems/sources;
- Health of the population (e.g., nutritional status, disease burden and previous outbreaks);
- Infrastructure (e.g., transport links, healthcare, public health infrastructure);
- Cultural practices and beliefs;
- Political environment (civil wars, terrorist attacks)
- **Scenario example 1:** For measles, outbreak spread depends upon factors like the current immunization coverage in the population, the capacity to quickly organize mass vaccination campaigns if coverage is low, general hygiene and access to health care, the existing surveillance capacity to rapidly detect and isolate cases, and population movement and behavior.
- **Scenario example 2:** For water contamination events by chemical agent, the risk of human intoxication will depend on factors like local water use practices, seasonality (cold or hot, rainy or dry), the flow of the water source, capacity to quickly broadcast risk and prevention messages to the public, and public acceptability of control measures.

Possible questions to ask

Are effective treatments and control measures available in the country?

- NOTE: Even if treatment and control measures exist somewhere in the world, answer “no” if they are not immediately and widely available in the specific setting where the event is occurring.

Is there an ongoing socio-political crisis that could hinder mobilization of resources or possible interventions?

Are the affected areas accessible to responders? (e.g. free of hostility/conflict)

Risk Characterization

Once the EBS unit has carried out the hazard, exposure, and context assessments, a level of risk should be assigned. This process is called risk characterization. For some units, risk characterization results in mathematical output from a quantitative model or comparison with an external standard value. But an equally acceptable process may result in a risk characterization based on the expert opinion of the EBS unit, with input from SMEs. Several tools have been developed to assist with the risk assessment and characterization process (e.g. WHO [manual for](#)

[the rapid risk assessment of acute public health events](#),¹ [ECDC Operational tool on rapid risk assessment](#),² Tripartite [Joint Risk Assessment Operational Tool](#)).³

Below we list two generic tools, a risk matrix and risk algorithm, that have been adapted from WHO and Africa CDC methodology, respectively. Countries are encouraged to explore and adapt the methodologies that fit best for them.

Risk matrix

This risk matrix combines estimates of the likelihood of event spread with estimates of the event consequences. As most acute health event risk assessments are qualitative, the categories used in the matrix are not based on numerical values but on broad descriptive definitions of likelihood and consequences (see [Figure 1](#) and [Tables 1–2](#)). When applying the matrix, the definitions of likelihood and consequence can be refined to fit with the national or intermediate-level context in each country.

- 1 WHO: Rapid risk assessment of acute public health events
- 2 Operational tool on rapid risk assessment methodology - ECDC 2019
- 3 Tripartite: Joint Risk Assessment tool

Figure 1. Risk characterization matrix

Consequences					
Almost certain					
Highly likely					
Likely					
Unlikely					
Very unlikely					
	Minimal	Minor	Moderate	Major	Severe

Likelihood

Table 1. Estimates of likelihood

Level	Definition
Almost certain	Is expected to occur in most circumstances (e.g., probability of 95% or more)
Highly likely	Will probably occur in most circumstances (e.g., a probability of between 70%–94%)
Likely	Will occur some of the time (e.g., a probability of between 30% and 69%)
Unlikely	Could occur some of the time (e.g., a probability of between 5% and 29%)
Very unlikely	Could occur under exceptional circumstances (e.g., a probability of less than 5%)

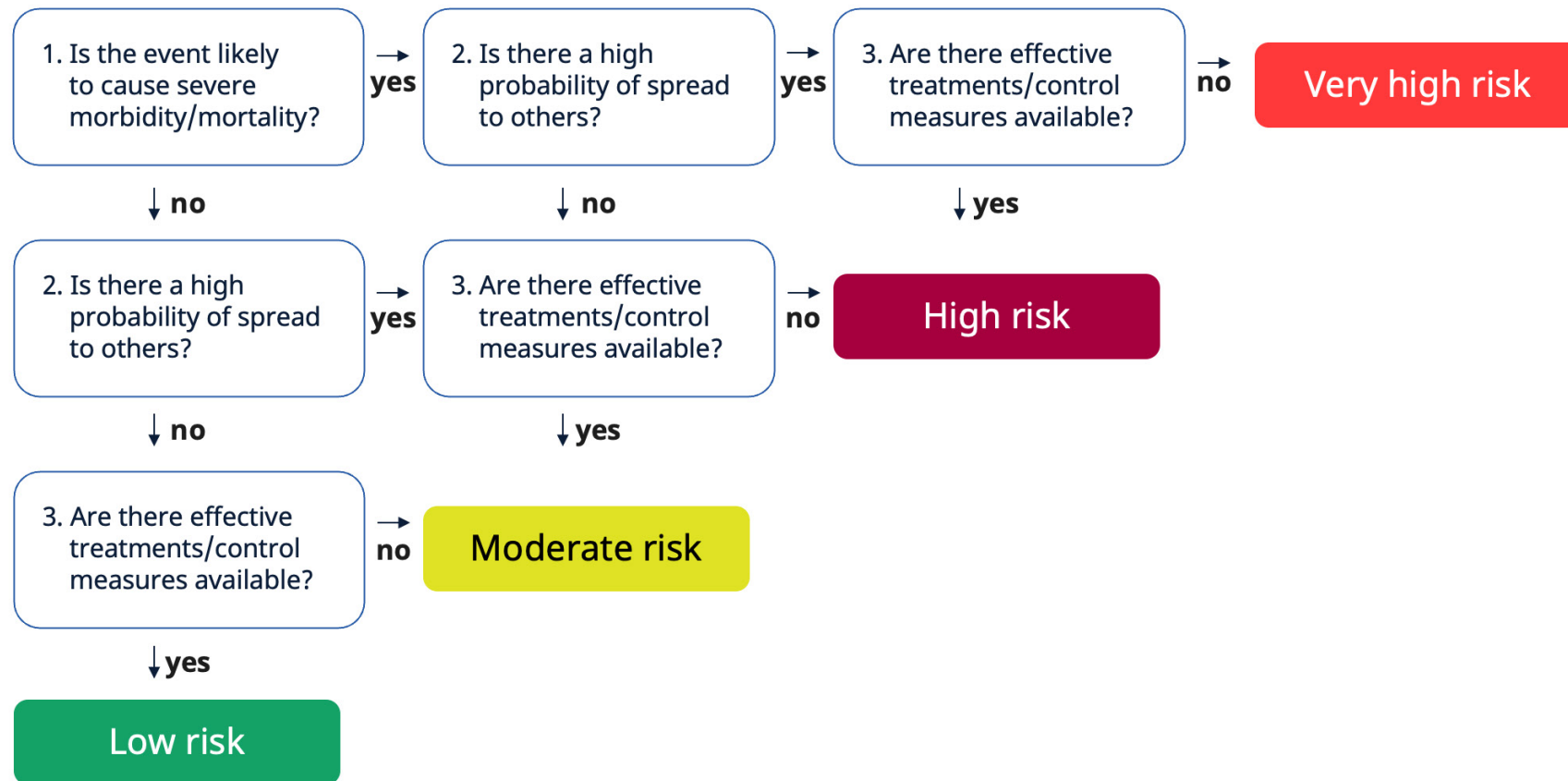
Table 2. Estimates of consequences

Level	Definition
Minimal	• Limited impact on the affected population • Little disruption to normal activities and services • Routine responses are adequate and there is no need to implement additional control measures • Few extra costs for authorities and stakeholders
Minor	• Minor impact for a small population or at-risk group • Limited disruption to normal activities and services • A small number of additional control measures will be needed that require minimal resources • Some increase in costs for authorities and stakeholders
Moderate	• Moderate impact as a large population or at-risk group is affected • Moderate disruption to normal activities and services • Some additional control measures will be needed and some of these require moderate resources to implement • Moderate increase in costs for authorities and stakeholders
Major	• Major impact for a small population or at-risk group • Major disruption to normal activities and services • A large number of additional control measures will be needed and some of these require significant resources to implement • Significant increase in costs for authorities and stakeholders
Severe	• Severe impact for a large population or at-risk group • Severe disruption to normal activities and services • A large number of additional control measures will be needed and most of these require significant resources to implement • Serious increase in costs for authorities and stakeholders

Risk algorithm

The below risk algorithm from the Africa CDC EBS framework is a series of questions that reflect upon the hazard, exposure, and context assessments and allow for a risk determination to be made based upon the responses to these questions.

Figure 2. Risk Assessment Algorithms



Note: If there are specific groups at increased risk of infection, consider performing separate risk assessment for each group. If in doubt, select higher risk answer.

The risk assessment team should decide how frequently the risk assessment should be updated. Usually, if there is an observed change that entails escalation, or de-escalation, of interventions, the risk assessment should be reviewed and updated.

Alert

Regardless of the source, once an event has been verified and the risk assessed, the responsible authorities should be alerted to respond to the event accordingly. This involves the immediate communication or notification of the event to the authorities designated for response and further action. The type of action taken will be dependent on the level assigned from the rapid risk assessment (Table 3, next page).

The most critical component of early warning and response systems is the response element. This has been widely covered in IDSR manuals (<https://apps.who.int/iris/handle/10665/112667>) and other WHO documents and will not be addressed in this document.

Table 3. Risk levels and recommended actions

Level	Recommended actions
Low risk	• Manage according to standard response protocols, routine control programs and regulation for event type. Possible EBS program actions include: • Listing on an event management system for monitoring until event is over or below defined alert threshold for endemic disease • Monitoring on weekly basis and repeating a risk assessment if situation changes
Moderate risk	• Manage according to standard response protocols, routine control programs and regulation for event type, but consideration should be given to enhancing existing surveillance/diagnostics, clinical management, and prevention efforts. Possible EBS program actions include: • Issuing a standard alert to authorities responsible for action • Including event on a daily or weekly event report for situational awareness • Listing on an event management system for monitoring until event is over or below defined alert threshold for endemic disease • Monitoring on a daily or weekly basis and repeating a risk assessment if situation changes
High risk	• Manage according to emergency response protocols established for event type. May be a need to establish command and control structures to coordinate response efforts. A range of additional control measures should be established in addition to conducting a field investigation, enhancing existing surveillance/diagnostics, clinical management, and prevention efforts. Possible EBS program actions include: • Issuing an immediate alert to senior management and all authorities responsible for action • Including event on a daily event report for situational awareness • Listing on an event management system for monitoring until event is over or below defined alert threshold for endemic disease • Monitoring daily and repeating a risk assessment if situation changes • Deploying response staff
Very high risk	• Manage according to emergency response protocols established for event with serious consequences. Immediate (within hours) establishment of command-and-control structures to coordinate response efforts. Implement immediate control measures in addition to enhancing existing surveillance/diagnostics, clinical management, and prevention efforts. Possible EBS program actions include: • Issuing an immediate alert to senior management and all authorities responsible for action regardless of routine working hours • Including event on a daily event report for situational awareness • Developing social media updates to inform general public • Listing on an event management system for monitoring until event is over or below defined alert threshold for endemic disease • Monitoring on an hourly/daily basis and repeating a risk assessment if situation changes • Deploying response staff • Soliciting external response support and establishing regional coordination